

THREADS: Visual Fundamentals

Cheatsheet: Race Conditions, Mutex, & Deadlocks

1. What is a Thread?

The Stack: Software → OS → Hardware.

CPU executes, RAM stores. The OS manages the relationship.

- **Multitasking vs. Threads:**

- *Multitasking:* Many apps (Browser, Music, VS Code) sharing CPU time.

- *Threads:* Many tasks inside one app (UI render, Spell-check, Autosave).

- **Timesharing:** CPU switches between threads so fast (milliseconds) it feels simultaneous.

Memory Hook: CPU = Referee, Threads = Players, RAM = The stadium.

2. Race Conditions

When two threads touch the same shared memory simultaneously.

- **The Dangerous Three Steps:** READ → MODIFY → WRITE.

- If a thread is interrupted between READ and WRITE, another thread reads the old, stale value.

Memory Hook: Two ATMs, same account, same second. Both approve a withdrawal; bank loses money.

3. Mutex (The Fix)

Mutual Exclusion: One thread at a time.

1. Thread requests lock.
2. If free, grab lock. If not, wait in queue.
3. Work is done.
4. Release lock.

Memory Hook: One bathroom key. Door locked = occupied, wait outside.

4. Deadlocks

The Circular Wait: Thread A holds Lock 1, wants Lock 2. Thread B holds Lock 2, wants Lock 1. Both freeze forever.

4 Conditions for Deadlock (All must be true):

1. **Mutual Exclusion:** Resource only held by one.
2. **Hold & Wait:** Thread holds one lock while waiting for another.
3. **No Preemption:** Cannot force a release.
4. **Circular Wait:** $A \rightarrow B, B \rightarrow A$ closed loop.

Memory Hook: Dining Philosophers. 5 people, 5 chopsticks. Everyone grabs left $\rightarrow$ nobody can eat.

5. How to Fix Deadlocks

- **Lock Ordering:** Always acquire locks in the same sequence.
- **Try-Lock + Timeout:** Give up after Xms, release held locks, and retry.
- **Deadlock Detection:** OS detects circular waits and kills a "victim" thread.
- **Message Passing:** Don't share state. Send messages instead (Go, Rust, Erlang).

6. Memory Hook Summary

Concept	Memory Hook
Thread	Referee spinning between players.
Timesharing	Chef flipping 5 dishes quickly; none burn.
Race Condition	Two ATMs, same second, bank loses money.
Mutex	One bathroom key; wait outside if locked.
Deadlock	5 philosophers, 5 chopsticks; nobody eats.